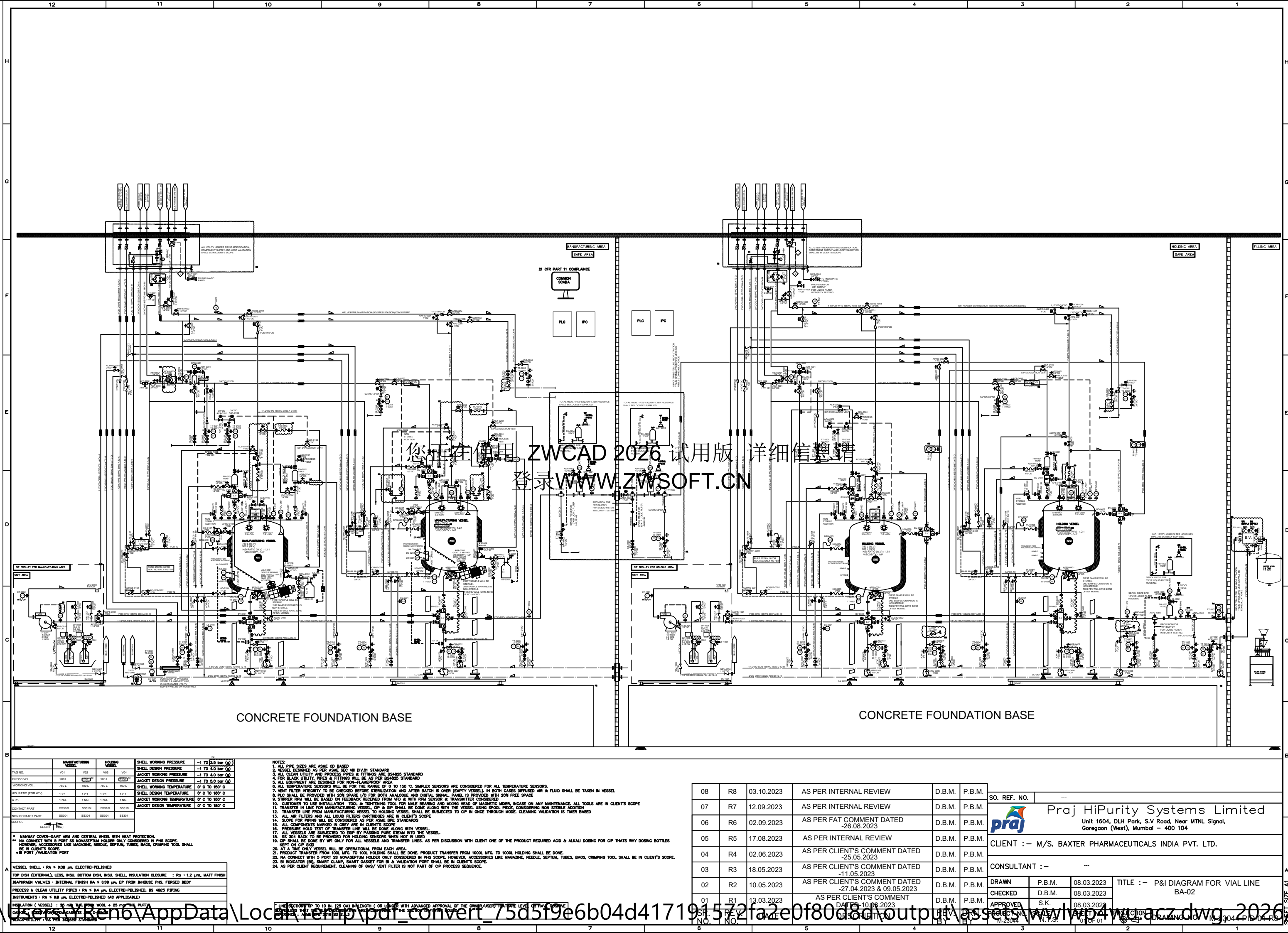




您正在使用 ZWCAD 2026 试用版 详细信息请
登录 WWW.ZWSOFT.CN

CONCRETE FOUNDATION BASE

CONCRETE FOUNDATION BASE

NO.	REV.	DATE	DESCRIPTION	BY	CHKD.	APPD.
08	R8	03.10.2023	AS PER INTERNAL REVIEW		D.B.M.	P.B.M.
07	R7	12.09.2023	AS PER INTERNAL REVIEW		D.B.M.	P.B.M.
06	R6	02.09.2023	AS PER FAT COMMENT DATED -26.08.2023		D.B.M.	P.B.M.
05	R5	17.08.2023	AS PER INTERNAL REVIEW		D.B.M.	P.B.M.
04	R4	02.06.2023	AS PER CLIENT'S COMMENT DATED -25.05.2023		D.B.M.	P.B.M.
03	R3	18.05.2023	AS PER CLIENT'S COMMENT DATED -11.05.2023		D.B.M.	P.B.M.
02	R2	10.05.2023	AS PER CLIENT'S COMMENT DATED -27.04.2023 & 09.05.2023		D.B.M.	P.B.M.
01	R1	13.03.2023	AS PER CLIENT'S COMMENT DATED -10.08.2023		D.B.M.	P.B.M.

SO. REF. NO. ---

Praj HiPurity Systems Limited
Unit 1604, DLH Park, S.V Road, Near MTNL Signal, Goregaon (West), Mumbai - 400 104

CLIENT :- M/S. BAXTER PHARMACEUTICALS INDIA PVT. LTD.

CONSULTANT :- ---

DRAWN P.B.M. 08.03.2023
CHECKED D.B.M. 08.03.2023
APPROVED S.K. 08.03.2023

TITLE :- P&I DIAGRAM FOR VIAL LINE BA-02

DRIVING NO. M-23044 N.P.S. 01 OF 01

您正在使用 ZWCAD 2026 试用版 详细信息请
登录 WWW.ZWSOFT.CN

CONCRETE FOUNDATION BASE

CONCRETE FOUNDATION BASE

NO.	REV.	DATE	DESCRIPTION	BY	CHKD.	APPD.
08	R8	03.10.2023	AS PER INTERNAL REVIEW		D.B.M.	P.B.M.
07	R7	12.09.2023	AS PER INTERNAL REVIEW		D.B.M.	P.B.M.
06	R6	02.09.2023	AS PER FAT COMMENT DATED -26.08.2023		D.B.M.	P.B.M.
05	R5	17.08.2023	AS PER INTERNAL REVIEW		D.B.M.	P.B.M.
04	R4	02.06.2023	AS PER CLIENT'S COMMENT DATED -25.05.2023		D.B.M.	P.B.M.
03	R3	18.05.2023	AS PER CLIENT'S COMMENT DATED -11.05.2023		D.B.M.	P.B.M.
02	R2	10.05.2023	AS PER CLIENT'S COMMENT DATED -27.04.2023 & 09.05.2023		D.B.M.	P.B.M.
01	R1	13.03.2023	AS PER CLIENT'S COMMENT DATED -10.08.2023		D.B.M.	P.B.M.

SO. REF. NO. ---

Praj HiPurity Systems Limited
Unit 1604, DLH Park, S.V Road, Near MTNL Signal, Goregaon (West), Mumbai - 400 104

CLIENT :- M/S. BAXTER PHARMACEUTICALS INDIA PVT. LTD.

CONSULTANT :- ---

DRAWN P.B.M. 08.03.2023
CHECKED D.B.M. 08.03.2023
APPROVED S.K. 08.03.2023

TITLE :- P&I DIAGRAM FOR VIAL LINE BA-02

DRIVING NO. M-23044 N.P.S. 01 OF 01

您正在使用 ZWCAD 2026 试用版 详细信息请
登录 WWW.ZWSOFT.CN

CONCRETE FOUNDATION BASE

CONCRETE FOUNDATION BASE

NO.	REV.	DATE	DESCRIPTION	BY	CHKD.	APPD.
08	R8	03.10.2023	AS PER INTERNAL REVIEW		D.B.M.	P.B.M.
07	R7	12.09.2023	AS PER INTERNAL REVIEW		D.B.M.	P.B.M.
06	R6	02.09.2023	AS PER FAT COMMENT DATED -26.08.2023		D.B.M.	P.B.M.
05	R5	17.08.2023	AS PER INTERNAL REVIEW		D.B.M.	P.B.M.
04	R4	02.06.2023	AS PER CLIENT'S COMMENT DATED -25.05.2023		D.B.M.	P.B.M.
03	R3	18.05.2023	AS PER CLIENT'S COMMENT DATED -11.05.2023		D.B.M.	P.B.M.
02	R2	10.05.2023	AS PER CLIENT'S COMMENT DATED -27.04.2023 & 09.05.2023		D.B.M.	P.B.M.
01	R1	13.03.2023	AS PER CLIENT'S COMMENT DATED -10.08.2023		D.B.M.	P.B.M.

SO. REF. NO. ---

Praj HiPurity Systems Limited
Unit 1604, DLH Park, S.V Road, Near MTNL Signal, Goregaon (West), Mumbai - 400 104

CLIENT :- M/S. BAXTER PHARMACEUTICALS INDIA PVT. LTD.

CONSULTANT :- ---

DRAWN P.B.M. 08.03.2023
CHECKED D.B.M. 08.03.2023
APPROVED S.K. 08.03.2023

TITLE :- P&I DIAGRAM FOR VIAL LINE BA-02

DRIVING NO. M-23044 N.P.S. 01 OF 01

This figure displays two detailed P&ID (Process and Instrumentation Diagram) drawings for a pharmaceutical manufacturing facility, specifically focusing on the vial line BA-02. The drawings are oriented horizontally, showing the flow from left to right through various processing stages.

The top half of each drawing shows the process vessels and piping, including a Manufacturing Vessel, Holding Vessel, and Filling Area. Key components include:

- Manufacturing Vessel:** A large cylindrical vessel equipped with internal stirrers, heating/cooling jackets, and multiple ports for utilities and product transfer.
- Holding Vessel:** A smaller cylindrical vessel used for holding the product before filling.
- Filling Area:** The final stage where the product is dispensed into vials.

The bottom half of each drawing illustrates the complex instrumentation network, featuring:

- Piping:** Extensive networks of pipes connecting vessels, tanks, and equipment, labeled with sizes and materials.
- Instruments:** Numerous symbols representing sensors (temperature, pressure, level), control valves, pumps, and actuators.
- Control System:** Integration with PLC (Programmable Logic Controller) and PC (Personal Computer) units for automated process control.
- Safety Features:** Indicated by specific symbols and labels such as "SAFE AREA" and "HOLDING AREA".

Below the main diagrams, there are several tables providing technical specifications and project details:

MANUFACTURING VESSEL	HOLDING VESSEL
TAG NO. V01	TAG NO. V02
GROSS VOL. 800 L	GROSS VOL. 800 L
WORKING VOL. 750 L	WORKING VOL. 750 L
HD. RATIO (FOR W/V) 1.2:1	HD. RATIO (FOR W/V) 1.2:1
QTY. 1 NO.	QTY. 1 NO.
CONTACT PART S0316L, S0316R, S0316L, S0316R	CONTACT PART S0304, S0304, S0304, S0304

SCOPE:-
* MAINWAY COVER-DRAFT ARM AND CENTRAL WHEEL WITH HEAT PROTECTION.
* NA CONNECT WITH 5 PORT SS NOVASEPTUM HOLDER ONLY CONSIDERED IN THIS SCOPE.
* HOWEVER ACCESSORIES LIKE MAGAZINE, NEEDLE, SEPTUM, TUBES, BAGS, CRIMPING TOOL SHALL BE IN CLIENT'S SCOPE.
* IF PORT / VALIDATION PORT

VESEL SHELL : RA ≤ 0.38 μm, ELECTRO-POLISHED
TOP DISH (EXTERNAL), LEGS, INSU. BOTTOM DISH, INSU. SHELL, INSULATION CLOSURE : Ra = 1.2 μm, MATT FINISH
DIAPHRAGM VALVES : INTERNAL FINISH RA ≤ 0.38 μm, EP FROM INHOUSE PHIS, FORGED BODY
PROCESS & CLEAN UTILITY PIPES : RA ≤ 0.4 μm, ELECTRO-POLISHED, 3G 4803 PIPING
INSTRUMENTS : RA ≤ 0.8 μm, ELECTRO-POLISHED GAS APPLICABLE
INSULATION (VESSEL) : AS PER THE DRAWING WOOL + 25 mm MIN. PURITY
HEATING/COOLING : AS PER THE DRAWING
HEAD OF UTILITY : AS PER JONES STANDARD

NOTES:

- ALL PIPE SIZES ARE ASME OD BASED
- VESSEL DESIGNED AS PER ASME SEC VIII DIV.01 STANDARD
- ALL CLEAN UTILITY AND PROCESS PIPES & FITTINGS ARE BS4825 STANDARD
- FOR BLACK UTILITY, PIPES & FITTINGS WILL BE AS PER BS4825 STANDARD
- ALL EQUIPMENT ARE DESIGNED FOR NON-FLAMMABLE AREA
- TEMPERATURE SENSORS WILL BE FOR THE RANGE OF 0 TO 150 °C. SIMPLEX SENSORS ARE CONSIDERED FOR ALL TEMPERATURE SENSORS.
- VENT FILTER INTEGRITY TO BE CHECKED BEFORE STERILIZATION AND AFTER BATCH IS OVER (EMPTY VESSEL). IN BOTH CASES DIFFUSED AIR & FLUID SHALL BE TAKEN IN VESSEL
- PLC SHALL BE PROVIDED WITH 20% SPARE I/O FOR BOTH ANALOGUE AND DIGITAL SIGNAL. PANEL IS PROVIDED WITH 20% FREE SPACE
- STIRrer RPM WILL BE BASED ON FEEDBACK RECEIVED FROM VFD & WITH RPM SENSOR & TRANSMITTER CONSIDERED
- CUSTOMER TO USE INSTALLATION TOOL & TIGHTENING TOOL, FOR MALE BEARING AND MIXING HEAD OF MAGNETIC MIXER, INCASE ON ANY MAINTENANCE. ALL TOOLS ARE IN CLIENT'S SCOPE
- TRANSFER IN LINE FOR MANUFACTURING VESSEL, CP & SP SHALL BE DONE ALONG WITH THE VESSEL USING SPOOL PIECE, CONSIDERING NON STERILE ADDITION
- TRANSFER LINE FROM MANUFACTURING VESSEL TO HOLDING VESSEL, SHALL BE SUBJECTED TO CP IN ONCE THROUGH MODE. CLEANING VALIDATION IS TIMER BASED
- ALL AIR FILTERS AND ALL LIQUID FILTERS CARTRIDGES ARE IN CLIENT'S SCOPE
- SLOPE FOR PIPING WILL BE CONSIDERED AS PER ASME PIPE STANDARDS
- ALL COMPONENTS MARKED IN GREY ARE IN CLIENT'S SCOPE
- PRESSURE HOLD TEST OF TRANSFER LINE WILL BE DONE ALONG WITH VESSEL
- ALL VESSELS ARE SUBJECTED TO ESD BY PASSING PURE STEAM INTO THE VESSEL
- SS 304 RACK TO BE PROVIDED FOR HOLDING SENSORS WHEN NOT IN USED
- CP SHALL BE DONE BY MS ONLY FOR ALL VESSELS AND TRANSFER LINES. AS PER DISCUSSION WITH CLIENT ONE OF THE PRODUCT REQUIRED ACID & ALKALI DOSING FOR CP THATS WHY DOSING BOTTLES KEPT ON CP SKD
- AT A TIME ONLY VESSEL WILL BE OPERATIONAL FROM EACH AREA
- PRODUCT TRANSFER FROM TOOL MFG. TO 100L HOLDING SHALL BE DONE. PRODUCT TRANSFER FROM 1000L MFG. TO 1000L HOLDING SHALL BE DONE.
- NA CONNECT WITH 5 PORT SS NOVASEPTUM HOLDER ONLY CONSIDERED IN THIS SCOPE. HOWEVER, ACCESSORIES LIKE MAGAZINE, NEEDLE, SEPTUM, TUBES, BAGS, CRIMPING TOOL SHALL BE IN CLIENT'S SCOPE.
- SI INDICATOR (SI), SMART CLAMP, SMART GASKET FOR IF & VALIDATION PORT SHALL BE IN CLIENT'S SCOPE
- AS PER CLIENT REQUIREMENT, CLEANING OF GAS/ VENT FILTER IS NOT PART OF CP PROCESS SEQUENCE.

PROJECT DETAILS:

REV	NO.	DATE	DESCRIPTION	BY	CHKD	APPD
08	R8	03.10.2023	AS PER INTERNAL REVIEW	D.B.M.	P.B.M.	
07	R7	12.09.2023	AS PER INTERNAL REVIEW	D.B.M.	P.B.M.	
06	R6	02.09.2023	AS PER FAT COMMENT DATED -26.08.2023	D.B.M.	P.B.M.	
05	R5	17.08.2023	AS PER INTERNAL REVIEW	D.B.M.	P.B.M.	
04	R4	02.06.2023	AS PER CLIENT'S COMMENT DATED -25.05.2023	D.B.M.	P.B.M.	
03	R3	18.05.2023	AS PER CLIENT'S COMMENT DATED -11.05.2023	D.B.M.	P.B.M.	
02	R2	10.05.2023	AS PER CLIENT'S COMMENT DATED -27.04.2023 & 09.05.2023	D.B.M.	P.B.M.	
01	R1	13.03.2023	AS PER CLIENT'S COMMENT DATED -10.08.2023	D.B.M.	P.B.M.	

CLIENT INFORMATION:

Praj HiPurity Systems Limited
Unit 1604, DLH Park, S.V Road, Near MTNL Signal, Goregaon (West), Mumbai - 400 104
CLIENT :- M/S. BAXTER PHARMACEUTICALS INDIA PVT. LTD.

CONSULTANT :-

DRAWN P.B.M. 08.03.2023
CHECKED D.B.M. 08.03.2023
APPROVED S.K. 08.03.2023
DESIGNED N.P.S. 08.03.2023

TITLE :- P&I DIAGRAM FOR VIAL LINE BA-02

DRAWING NO. M-304-RS-01